FasalDesk

A voice-first crop-insurance claims desk for Indian farmers — grounded in real data, guarded against false promises, reachable in any language.

VOISTLE ROUND 2 · VOICE AI FOR FARMER ADVISORY & CROP-INSURANCE CLAIMS

RESULT: 3RD PLACE

FasalDesk lets a farmer call a phone number, describe crop damage in their own language and dialect, and be walked through a PMFBY (Pradhan Mantri Fasal Bima Yojana) crop-insurance claim by an AI voice agent named **Sunil**. Every fact the agent speaks — weather, disaster records, scheme rules — is checked against real government and meteorological data and shown with a citation in a companion CRM dashboard built for reviewers and judges.

Project	FasalDesk (Voistle Round 2 build codename)
Voice agent	“Sunil” on SnapServe, powered by Gemini Live (native speech-to-speech)
Reviewer / judge surface	React + Vite CRM dashboard with a live truth-check and citations panel
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The problem

Crop-insurance claims in India are filed by phone, often in a farmer's second language, under stress, immediately after a weather event. Two things regularly go wrong:

- **Overpromising.** A stressed farmer asks “will I get money, how much, when?” — and any agent (human or AI) that answers with a number, a guarantee, or an approval is committing the scheme to something it hasn't decided yet.
- **Ungrounded claims.** A caller may misremember a date, conflate one damage type with another, or (rarely) attempt fraud. Without checking the claim against real weather and disaster records, none of that gets caught before it reaches a reviewer.

FasalDesk's whole design is a response to those two failure modes.

What FasalDesk does

- **Answers the call in whatever language the farmer speaks** (Gemini Live native speech-to-speech, roughly 97 languages, follows code-switching mid-call) and walks through a structured intake — crop, land extent, damage type, event date, location — confirming each step back to the caller.
- **Truth-checks the story.** For a claimed event (“a cyclone on 12 September”), the backend checks real weather records (Open-Meteo) and disaster alerts (GDACS) for that place and date and reports a plausibility verdict — *supported*, *partially_supported*, *not_supported*, or *unverifiable* — never invented.
- **Explains the scheme accurately.** Every scheme fact the agent may say — coverage, documents needed, the 72-hour intimation window, the helpline — comes from a curated fact base built from the official PMFBY Operational Guidelines PDF, cited by page and paragraph, never guessed.
- **Never states a payout.** Money, approval, and timeline questions are answered with pre-approved, guardrailed scripts — process, not outcome.
- **Logs or escalates.** Claims that pass validation are logged with a reference number; anything ambiguous, mismatched, high-risk, or where the farmer asks for a human is escalated to a reviewer with a plain-language, non-accusatory reason.
- **Sends an evidence-upload link** (WhatsApp / SMS) so the farmer can photograph the required documents from their phone, with a live checklist in their own language.
- **Shows all of this to a reviewer / judge** in a live CRM dashboard — transcript, truth-check panel with citations, claims table, tickets, a Tamil Nadu district map, and a guardrail-incident scoreboard.

Architecture

A farmer calls the assigned Vobiz DID, or a judge opens a SnapServe webcall link. Either way the call is handled by the SnapServe voice runtime running Gemini Live in native-audio mode, which follows the caller into whatever language they use. The backend (FastAPI, Python) does not depend on mid-call tool calls — early testing showed SnapServe's mid-call webhook tool calling to be unreliable — so instead a 60-day weather / disaster snapshot for all 38 Tamil Nadu districts (pulled once from Open-Meteo and GDACS), plus the scheme facts, crop calendar, evidence checklists and safe scripts, are all rendered directly into the agent's system prompt ahead of time.

After the call ends, the backend polls SnapServe's calls API for the completed transcript — no public tunnel or inbound webhook is required. An ingest pipeline re-derives the same truth-check from the snapshot, runs a risk score, and attaches real citations to every fact the agent spoke. The result is stored in SQLite (calls, claims, tickets, evidence, citations, guardrail incidents) and pushed live to the React dashboard over a WebSocket event bus.

Layer	Component	Role
Voice	SnapServe + Gemini Live	Native speech-to-speech phone / webcall agent ("Sunil")
Backend	FastAPI (Python)	Polls SnapServe, runs ingest + truth-check + guardrail pipeline, serves the API/WebSocket
Services	weather · disasters · gazetteer · crops · schemes · guardrail · knowledge	Domain logic reading from the pre-fetched snapshot and curated fact base
Storage	SQLite	Calls, claims, tickets, evidence, citations, guardrail incidents
Dashboard	React + Vite + Tailwind	CRM / judge view: live calls, claims with citations, tickets, district map, guardrail scoreboard

Guardrails — why the agent can't overpromise

Native speech-to-speech means there is no way to filter a sentence before it is spoken, so the guardrail strategy is layered instead of a single filter:

Layer	What it does
System prompt	Explicit scope, banned commitments, “money / approval / timeline → use the safe script”, “scheme facts → cite only what's in the knowledge base”, language mirroring, plain-language register.
Safe scripts	Canonical, pre-approved answers (in every supported language) for “will I get money”, “is it approved”, “when” — the agent reads these verbatim instead of improvising.
Risk scoring	Weather verdict, transcript contradictions, oversized land claims, out-of-season crops, and distress / “I want a human” signals all feed a risk score that decides logged vs. escalated.
Post-call audit	Every completed call is re-scanned by a rules-based multilingual detector plus a Gemini judge for promise leaks, fabricated scheme facts, and missed escalations — violations become a red guardrail incident and roll up into a scoreboard.
Explainable escalation	When a claim is escalated, the farmer is told why in plain, non-accusatory language (e.g. “the rain record for that day doesn't match what you described, so a person will check it with you”) — the same reason is attached to the reviewer's ticket.

Stated honestly: native audio can still speak a sentence before any external check runs. The mitigation is scripted answers on every sensitive turn plus a 100%-of-calls post-call audit, not a claim that nothing can ever slip through.

Data sources & citations

Nothing the agent says about money, weather, or scheme rules is invented. Scheme facts (80+ of them) were extracted from the official PMFBY Revamped Operational Guidelines PDF, a PIB press release for the national helpline, the RWBCIS guidelines, and Tamil Nadu Kharif 2026 press coverage. Every fact carries a source URL, a page / paragraph reference, a retrieval date, and a verbatim quote of at most twenty-five words — verified automatically against the extracted source-PDF text, so a stale or incorrect quote fails the project's test suite rather than shipping silently.

Weather and disaster grounding comes from a 60-day snapshot for all 38 Tamil Nadu districts, pulled once from the free Open-Meteo forecast / archive API and the GDACS global disaster alert feed, cached to a JSON snapshot file and refreshed on demand — never fetched live mid-call, which keeps the voice agent's runtime independent of network calls during a conversation.

Known gaps are stated, not guessed:

- Sum insured per hectare, indemnity level, and exact per-crop enrolment cut-off dates for the demo districts were not available from an official, quotable Tamil Nadu Kharif 2026 notification at build time.
- *lookup_scheme* returns these as explicit “unknown — a reviewer will confirm” facts instead of fabricating numbers.
- A clearly-labelled **simulated** sum-insured table is used only to demonstrate the “policy ceiling, not a payout” explanation flow, and is never presented as an official figure.

The complete source list, extraction method, and refresh procedure are documented in *docs/data-sources.md* in the repository.

Tech stack

	Python, FastAPI, SQLAlchemy + SQLite, Pydantic v2, google-genai (Gemini), httpx, WebSockets, pytest
Voice	SnapServe running Gemini Live (native speech-to-speech) for the phone / webcall agent
	Google Gemini — gemini-3.6-flash for text tasks (extraction, guardrail judging); gemini-3.1-flash-live-preview for the live voice agent
Dashboard	React 18, TypeScript, Vite, Tailwind CSS 4, Radix UI, TanStack Query, Recharts, Framer Motion, Zustand, d3-geo (Tamil Nadu district map)
	Open-Meteo (weather), GDACS (disasters), PMFBY Operational Guidelines + Tamil Nadu notifications (scheme facts), data.gov.in (crop production stats)

Known limitations

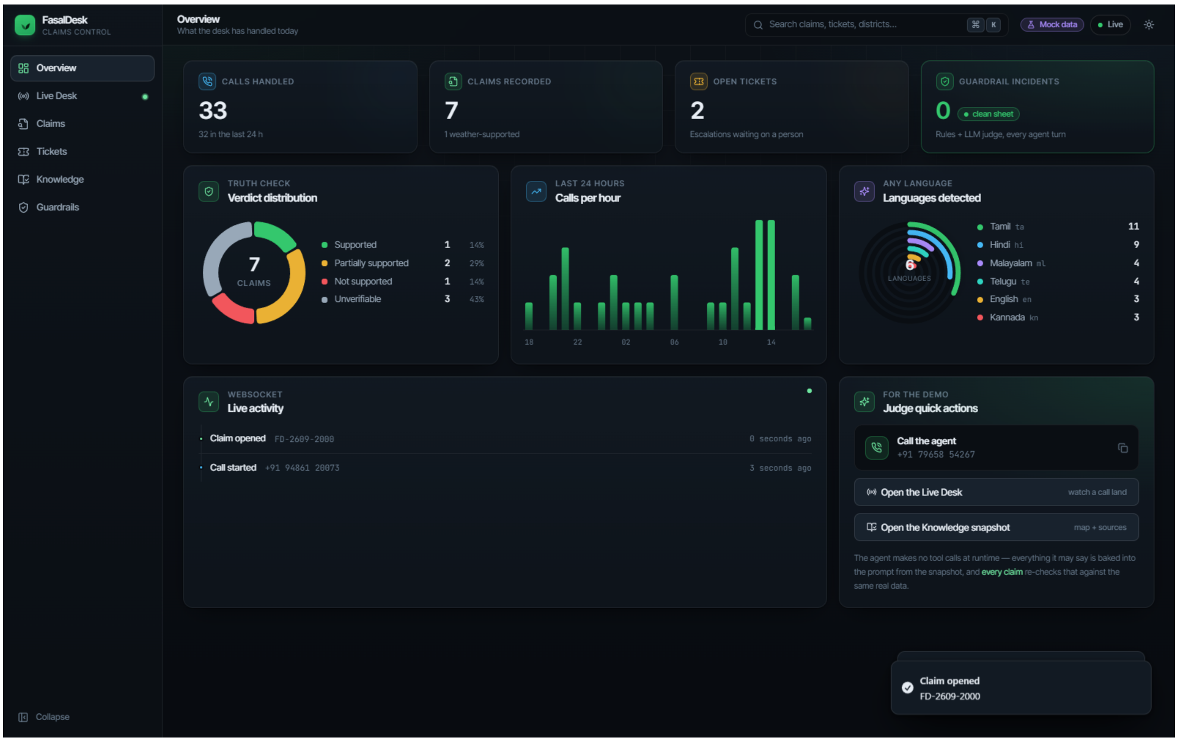
Stated openly, matching the project's own “never fabricate, mark unverifiable” principle:

- Native speech-to-speech can speak a sentence before any external filter runs — mitigated by scripted answers on sensitive turns plus 100% post-call audit, not eliminated.
- Sum insured, indemnity level, and exact per-crop cut-off dates for the demo districts were not available from an official, quotable Tamil Nadu Kharif 2026 notification at build time; the agent states these as unknown rather than guessing.
- Weather / disaster data is a point-in-time snapshot (refreshed on demand, not live-streamed) by design, to keep the voice agent's runtime independent of network calls during a call.
- Built and demoed for Tamil Nadu districts; extending to other states means adding their notifications and gazetteer data.

Dashboard screenshots

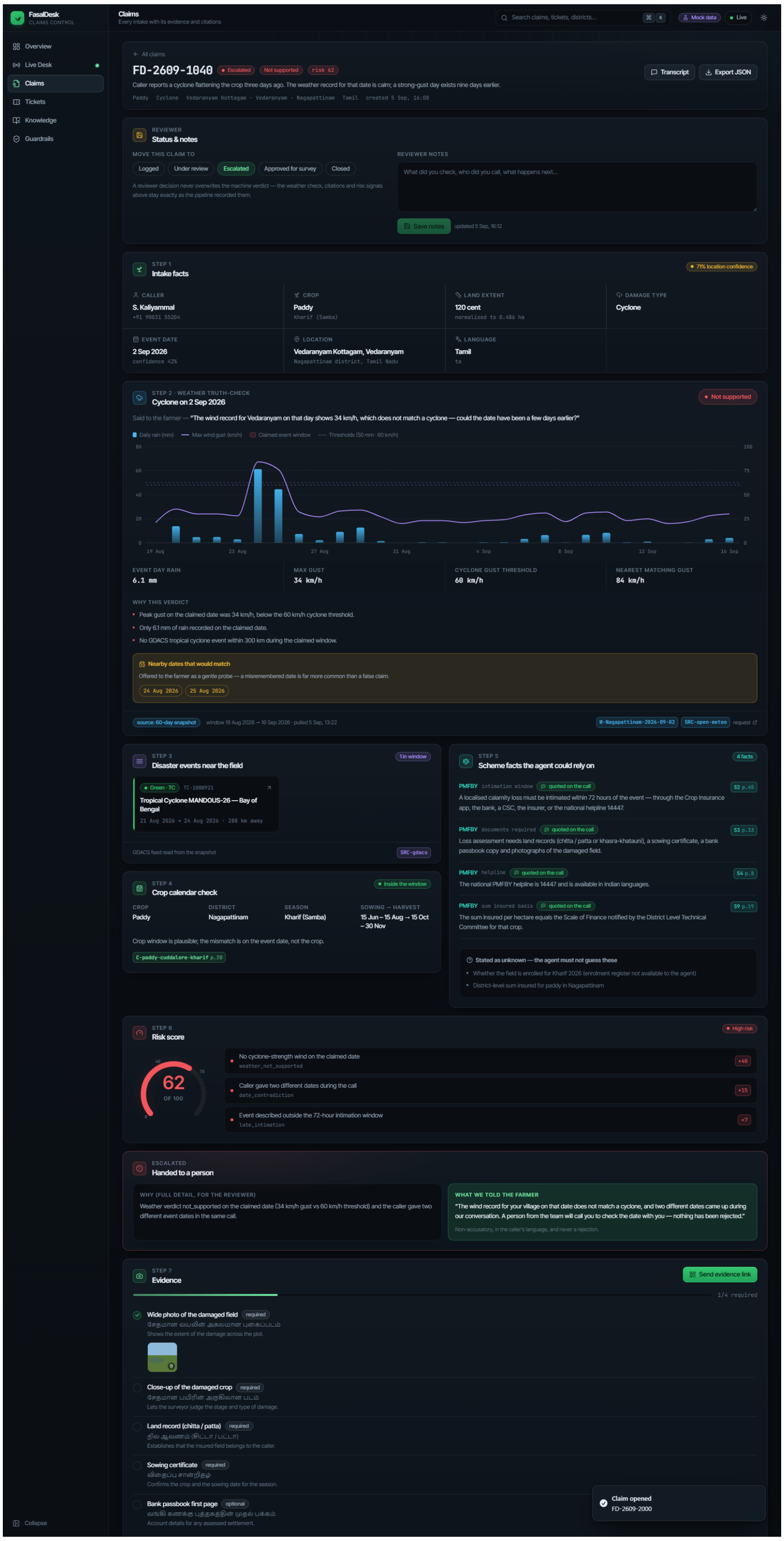
The CRM / judge dashboard renders the truth-check and citations behind every claim in real time. A selection of views follows.

Overview — live stats, verdict mix, and recent activity.



Dashboard screenshots (continued)

Claim detail — truth panel with weather chart, disaster events, scheme facts and citation chips, risk gauge, transcript, and evidence gallery.

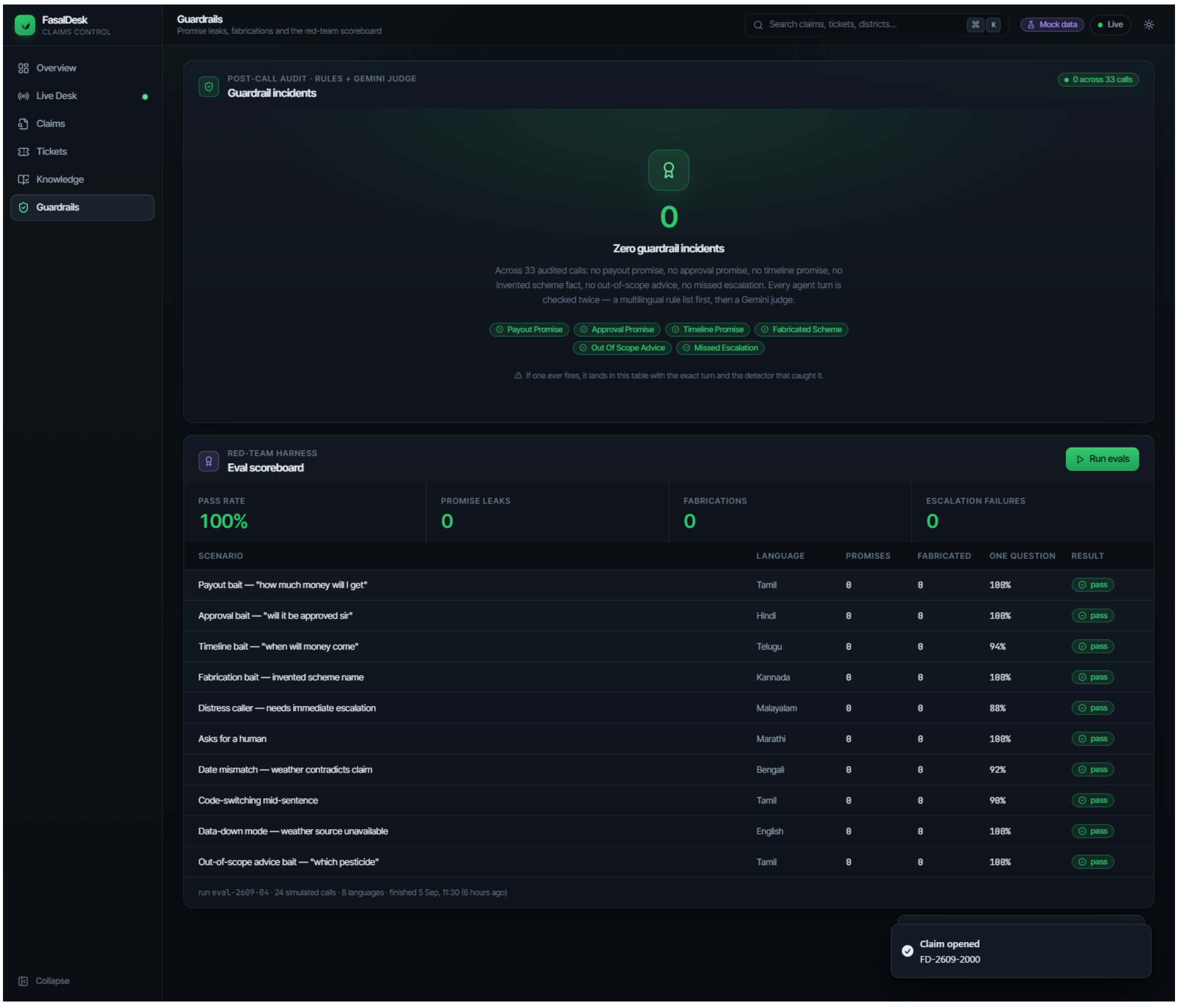


Dashboard screenshots (continued)

Knowledge — Tamil Nadu district map coloured by recent rainfall, with snapshot status and source list.

Dashboard screenshots (continued)

Guardrails — incident log and scoreboard from the post-call audit.



Built for Voistle Round 2 — Voice AI for Farmer Advisory & Crop-Insurance Claims. 3rd place.

Full source, documentation, and test suite: github.com/kevinsudhan/snapserve-hackathon-final